Order Scandentia, Family Tupaiidae

Tree shrews (*Anathana* 175-200 mm; *Dendrogale* 35-55 g; *Ptilocercus* 23-60 g; *Tupaia* 30-320 g; *Urogale* 350 g) are arboreal omnivores from tropical rainforests of southeast Asia. *Tupaia* males may be larger than females (4626). Ovulation may be induced (*Tupaia* 2011, 4474, 6890, 9917). The labyrinthine, bidiscoidal placenta has been described variously as hemochorial, endotheliochorial, or epitheliochorial, depending upon fetal age, and it has a superficial, bilateral, antimesometrial implantation at areas of the bicornuate uterus that are free of endometrial glands (231, 1491, 4764, 6087, 6622, 6623, 6890, 7179, 7180, 9684, 9685, 10310, 11111, 11256). Placentophagia occurs (*Tupaia* 10281) as does a postpartum, occasionally prepartum, estrus (*Tupaia* 3709, 4474, 4626, 6087, 6890, 7986, 9682, 9685, 9917, 10224, 10281, 10482, 11164; *Urogale* 2109). Delayed implantation may occur (6087, 6622, 6890; cf 9684). *Tupaia* young are suckled once every 48 hours (6889, 6892, 7986, 9682).

Tupaia neonates are naked and open their eyes at 13-25 d (398, 3706, 6889, 7161, 7535, 7986, 9217, 9917, 10223, 10280, 10281). Females menstruate (2011; cf 11164) and have 2-6 mammae (1990, 2313, 6678, 7161, 10279, 10281, 11899). Males have prepenial testes, seminal vesicles, and an ampullary gland (1491, 1990, 2310, 6890, 11256, 11899). *Anathana* males have abdominal testes (11256). Details of anatomy (136, 2777, 6311, 10482, 11256, 11899), endocrinology (1989, 7179, 9685), spermatogenesis (1591), and milk composition (2209, 6890, 9682, 11716b) are available.

NEONATAL MASS		
Tupaia glis	r10-12 g; n5	8798
	11.0-19.0 g	10223
	11.2 g; n8	10281
	12.6 g; suckled once	10280
	m: 12.9 g; r9.1-19.3	9682
	m: 14.5 g; n10	9917
	15 g; with ~6 g milk; r~5-20; n8	6889, 6890
Tupaia montana	11.0 g; r8.0-15.7; n3	10224
Urogale everetti	19.15 g; n1, 2 d	2109
	20.8 g; n1	2109
NEONATAL SIZE		
Tupaia glis	79 mm; n8	10281
	100-140 mm	10223
Urogale everetti	130 mm; 2 d	2109
WEANING MASS		
Tupaia spp	80 g; r~50-110; n8	6889
Tupaia glis	~100 g	9682
LITTER SIZE		
Anathana wroughtoni	5 emb; n1 f	11256
Tupaia glis	1.83 emb; mode 2; r1-2; n6 f	4626
	1.9; r1-2; n10 lt	5581
	mode 2; r1-4	11344
	2 emb	6201
	2 emb; n1 f	6678
	2 emb ea; n2 f	2313
	2 yg ea; n4 lt, 1 f	398
	2 yg ea; n9 lt	4474
	2.0; n5 lt	4406
	2.0 yg; r1-3; n57 lt	7535
	2.09 live yg; mode 2; r1-3; n21 lt	6890
	2.23; r1-4; mode 2; n56 lt	9682
	2.25 emb; r1-3; n4 f	6678
	2.37; r2-3; n8 lt	3706
	2.38; r1-3; n16 lt	10281
	2.5 yg; n54 lt	9683
	2.7; r2-3; n3 lt	7986
	2.83; r2-3; n6 lt	3706
	3 emb; n1 f	10824
	3-5 yg	4813a
Tupaia javanica	max 2	4764
	4-5 cl; 2 emb	10482

Tupaia minor	1-3	2209
	1.9 yg; r1-2; n8 lt	7161
	2 emb ea; n4 f	2313
	2; n48 lt	4406
Tupaia montana	r1-3 yg; n19	10224
Tupaia nicobarica	1 emb; n1 f	6678
Tupaia tana	1-3	2209
	2 emb ea; n3 f	2313
	2.5 yg; r2-3; n2 lt	9217
	4 emb; n1 f	6678
Urogale everetti	1.67; r1-2; n3 lt	2109
	1.86 yg; r1-2; n7 lt, 3 f	11612
SEXUAL MATURITY		
Tupaia glis	f: 1st birth: 150-300 d	9683
	f: 1st birth: 6 mo	10281
	f: 1st preg: 130 g	4406
	m: fertile: 50% 114.5 g, 95% 190.6 g	4406
	m: sperm: 90 d	1989
	m: 5 mo	398
	m: 6 mo	10281
	f/m: 90 d	5591
	f/m: 90-100 d	9917
	f/m: est 3-4 mo	9913
	f/m: 5-6 mo	7535
Tupaia minor	f: 1st preg: 46 g	4406
	m: fertile: 50% 16.3 g, 95% 449.7 g	4406
CYCLE LENGTH		
Tupaia glis	estrus: 2-4 h	2011
	10-12 d or 18-22 d	2011, 10223
Tupaia montana	r9-12 d; n16; intercopulation interval	10224
GESTATION		
Tupaia glis	implant: 7 d	9684
	43-45 d	9917
	43-46 d	10281
	43-46 d; n7 lt	4474
	43.2 d	9682
	44 d; n1	398
	44 d; r40-47	6087
	45 d, r40-52; n3	3706, 3709
	~45 d	7535
	est 46-50 d	10824
	46-50 d	4626
	~47 d; mode 49; r41-50; n39	2011, 10223
Tupaia montana	45-53 d; n20 lt	10224
Urogale everetti	~56 d	11612
LACTATION		
Tupaia glis	leave nest: 23 d	3706
	leave nestbox: 32 d	781a
	den emergence: 33 d	6890, 6892
	wean: few d after den emergence	6892
	solid food: 35 d	398
	wean: ~30 d	9682
	wean: ~30 d	11344
	wean: 6 wk	7535
Tupaia tana	solid food: 4 wk	9217
	solid food: 31 d	10223
	attempt nurse: 53 d	10223
Urogale everetti	den emergence: ~1 mo	2109
INTERLITTER INTERVAL		
Tupaia glis	44 d; mode 43-45; r41-120	9682, 9683, 9685
	45 d	11344
	min 45 d	6890, 6892
	45.3 d; r43-47	7986
	mode 49 d; r41-50	2011

SEASONALITY

Tupaia glis	testes: large Feb, small Sept; lab	1990
	mate: yr rd; lab	2011
	mate: Jan-May; captive	4626
	preg f: end Apr-mid Aug; n2/10; N Borneo, Indonesia	2313
	preg f: Aug; n1; Phu-Guy, Vietnam	10824
	birth: yr rd; Malaysia	7161
	lact/preg f: Jan-June, none July-Aug; Malaysia	6200
	repro: est yr rd; captive	4474
	repro: Feb-June; Penany I, Malaysia	6201
Tupaia minor	birth: yr rd, peak May-July; Malaysia	7161
Tupaia montana	birth: yr rd; captive	10224
Tupaia tana	preg f: July-Aug, Nov; no Feb-June data; Java, Banka, Indonesia	12120
	preg f: mid-end July; n3/15; N Borneo, Indonesia	2313
Urogale everetti	birth: Aug, Oct, Dec; n1 pr; zoo	2109